

Does Identity Mapping Really Help in ResNet?

Yoo-Kyung Lee^{1*}, Dong-Hwan Lee^{1*}, Jae-Hun Choi^{1,2}, Jang-Hee Yoo^{1,2}, Seung-Ik Lee^{1,2**}

¹University of Science and Technology

²Electronics and Telecommunications Research Institute

Daejeon, Korea

{lyk, ehgls, jhchoi, jhy, the_silee}@etri.re.kr

Abstract—ResNet proposed residual learning using skip connections that perform identity mapping. However, as shown in the study [1], identity mappings for skip connections may not be the optimal or at least not the only way to get a good performance in ResNet. In this paper, we explore various types of skip connections, including shortcuts with or without identity mapping, to see their effects on the performance of ResNet and provide a comparison of the various methods. Our extensive experiments suggest that the use of identity shortcuts is the most appropriate than other methods in terms of model simplicity and performance.

Keywords—skip connection, identity mapping, deep residual networks

I. INTRODUCTION

ResNet [1] was proposed to address the problem of performance as deep neural networks get deeper. Following ResNet's success in utilizing skip connections, models such as DenseNet [3] and FishNet [4] emerged and showed a great performance as well. In ResNet [1], the authors argue that the concept of identity mapping is crucial. However, not all shortcuts in ResNet are identity mapping. Specifically when shortcuts go across feature maps of two different sizes, the feature map size is halved and the channel increases. In that case, the shortcuts employ either 1x1 convolutions or zero-entry padding is utilized to fit the dimension change, thus practically breaking the ResNet's fundamental idea of identity mapping in skip connections. In other words, the identity mapping is no longer retained when skip connections use 1x1 convolutions or zero-entry padding. This naturally raises the question of whether these non-identity shortcut connections are an optimal choice considering the fact that identity mapping is the key concept in ResNet. In this paper, we conduct extensive experiments on various types of shortcut connections, with or without identity mappings to empirically demonstrate if the non-identity shortcuts can be replaced and if so, to find out alternatives that can show a better performance.

II. RELATED WORK

A. Skip Connection

Skip connection [5]–[7] is one of the important techniques for enabling performance improvement in deep layered models. They convey information to non-neighboring layers by

*These authors contributed equally to this work

**Corresponding author.

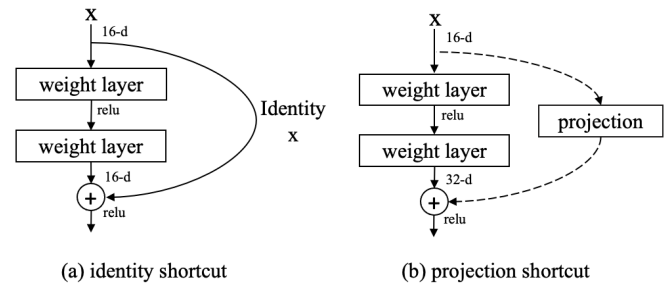


Fig. 1. Structure of shortcuts with an example of 16-dimensional input x . (a) Identity shortcut. (b) Projection shortcut. Projection shortcuts are for increasing dimensions.

skipping one or more layers and are also known to help avoid the vanishing gradient problem often observed in deeper neural networks. Specifically, skip connections can provide an alternative path through which gradients are passed backward. This contributes to improved performance and based on these advantages, there are various models that use skip connection [1], [3], [4], [8]–[11].

B. Identity Mapping

Identity mapping is to ensure that the equality of the input and output values of a layer is readily obtained. It is done by passing input data x directly to the layer's output so that the input information is not easily lost. In the residual network, this is implemented as shown in Fig.1 (a).

He. et al. [2] compared the original identity skip connections proposed by [1] to shortcuts with various transforms such as scaling, gating, and 1x1 convolutions. They found that using the original identity mapping scheme as the skip connections shows the best performance. However, they haven't dealt with the other type of shortcuts, i.e., non-identity shortcuts in ResNet such as 1x1 projection shortcuts or zero-padding shortcuts, which go across feature maps of different sizes.

Differently from [2], we put more focus on these non-identity mappings to find out if these mappings can be further improved by replacing the non-identity mappings with more identity-preserving mappings. In addition, we also provide comparisons on various shortcuts with and without identity mapping.

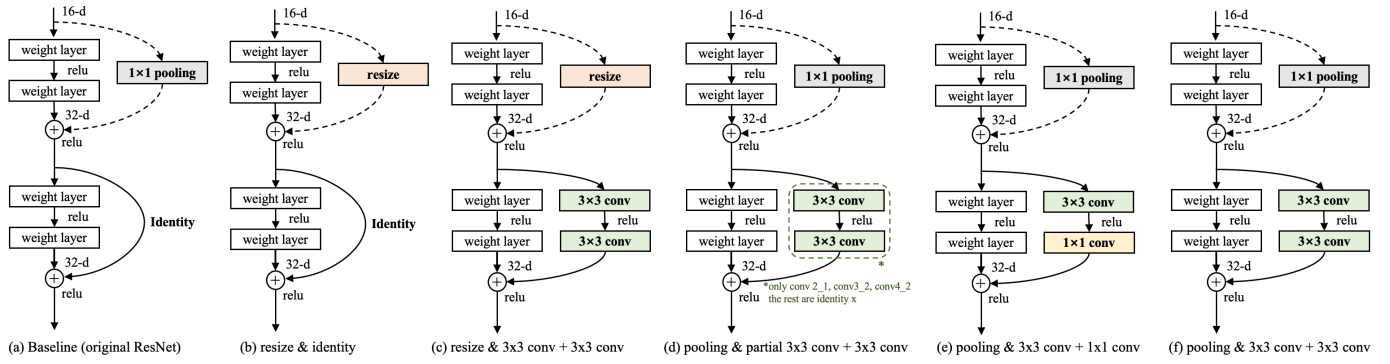


Fig. 2. Various shortcut connection replacements. Solid lines are when input and output dimensions are the same, and dotted lines are when the dimension increases. We do not draw the BN layers for simplicity.

TABLE I
ARCHITECTURES FOR CIFAR-10. DOWNSAMPLING AT CONV3_1 AND CONV4_1 WITH A STRIDE OF 2.

layer name	output size	20-layer	32-layer	44-layer	110-layer
conv1	32x32	3x3, 16, stride 1			
conv2_x	32x32	$\begin{bmatrix} 3 \times 3, 16 \\ 3 \times 3, 16 \end{bmatrix} \times 3$	$\begin{bmatrix} 3 \times 3, 16 \\ 3 \times 3, 16 \end{bmatrix} \times 5$	$\begin{bmatrix} 3 \times 3, 16 \\ 3 \times 3, 16 \end{bmatrix} \times 7$	$\begin{bmatrix} 3 \times 3, 16 \\ 3 \times 3, 16 \end{bmatrix} \times 18$
conv3_x	16x16	$\begin{bmatrix} 3 \times 3, 32 \\ 3 \times 3, 32 \end{bmatrix} \times 3$	$\begin{bmatrix} 3 \times 3, 32 \\ 3 \times 3, 32 \end{bmatrix} \times 5$	$\begin{bmatrix} 3 \times 3, 32 \\ 3 \times 3, 32 \end{bmatrix} \times 7$	$\begin{bmatrix} 3 \times 3, 32 \\ 3 \times 3, 32 \end{bmatrix} \times 18$
conv4_x	8x8	$\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 64 \end{bmatrix} \times 3$	$\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 64 \end{bmatrix} \times 5$	$\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 64 \end{bmatrix} \times 7$	$\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 32 \end{bmatrix} \times 18$
	1x1	average pool, 10-d fc, linear			

III. METHODS

The questions to be addressed are twofold: 1) Can the *non-identity shortcuts* in ResNet be further improved? 2) Are *identity shortcuts* essential? In other words, can't they be replaced with non-identity mappings? To address these two questions, we propose several replacements for the identity and non-identity shortcuts as shown in Fig. 2.

As for the first question, we propose to use *resize* instead of 1x1 pooling (zero-padding shortcuts) as shown in Fig. 2 (b) & (c). The 1x1 pooling (with stride 2 and zero padding for increased channels and downsampling) used in ResNet can be considered to have 75% loss of information because of skipping values. On the other hand, the proposed way of resizing feature maps take surrounding values into account as well when creating new values and therefore no pixels or features are skipped or ignored. This can be viewed as maintaining the identity concept much more than the 1x1 pooling way of the baseline proposed in ResNet with no additional parameters.

As for the second question, we propose several different types of replacements for the identity-preserving shortcuts and see if the connections without identity mapping yield any performance improvements. To destroy the identity, two convolution layers are employed in such a way that they destroy identity in an increasing order from Fig. 2 (d) to (f). Fig. 2 (d) applies two layers of 3x3 convolutions only to conv2_1, conv3_2, conv4_2, while all the other layers retain identity mapping. This configuration partially breaks

the identity mappings. Fig. 2 (e) and (f) apply two convolutional layers to all the identity shortcuts in ResNet. The 1x1 convolution of stride size 1 in Fig. 2 (e) can be regarded as a less distorting identity than 3x3 convolutions. For (d)-(f), zero-padding shortcuts (1x1 pooling) are used for downsampling and increasing channels just as in the baseline of Fig. 2 (a).

IV. EXPERIMENTS

A. Datasets and Model Configuration

We use CIFAR-10 which consists of 10 classes with 50k training images and 10k testing images, and CIFAR-100 of 100 classes with 50k images for training and 10k images for testing. Our model is constructed following the configuration in [1] by converting the baseline code¹ into Pytorch. The models for CIFAR-10 and CIFAR-100 are basically the same except for the last fully connected layer for a different number of classes (Table I).

B. Implementation Details

We use an SGD optimizer with a momentum of 0.9 and a weight decay of 0.0001. We adopt Pytorch's He weight initialization [12] and batch normalization [13] without dropout. We train our models on four GPUs with a mini-batches of size 128. As in the original ResNet, the learning rate is set to 0.01 up to 400 iterations and 0.1 thereafter until it is decreased again by 10 at 32k and 48k iterations. Training continues until 64k iterations. We augment the data by random cropping with

¹ResNet at <https://github.com/kuangliu/pytorch-cifar/blob/master/models>

TABLE II
CLASSIFICATION ERROR (%) ON CIFAR10. WE RUN IT 5 TIMES AND SHOW “MEAN±STD”. “FAIL” MEANS THE TEST ERROR IS OVER 50%.

Method	ResNet20	ResNet32	ResNet44	ResNet110
(a) Baseline	8.26±0.18	7.48±0.08	7.08±0.16	6.28±0.24
(b) Resize & identity	8.20±0.12	7.18±0.09	7.31±0.24	6.16±0.14
(c) Resize & 3x3 conv + 3x3 conv	7.66±0.15	8.48±0.30	9.93±0.33	fail
(d) Pooling & partial 3x3 conv + 3x3 conv	7.69±0.28	7.46±0.21	7.00±0.03	6.50±0.22
(e) Pooling & 3x3 conv + 1x1 conv	8.00±0.17	8.46±0.12	9.43±0.18	fail
(f) Pooling & 3x3 conv + 3x3 conv	7.70±0.15	8.57±0.20	10.5±0.23	fail

TABLE III
CLASSIFICATION ERROR (%) ON CIFAR100. WE RUN IT 5 TIMES AND SHOW “MEAN±STD”. “FAIL” MEANS THE TEST ERROR IS OVER 50%.

Method	ResNet20	ResNet32	ResNet44	ResNet110
(a) Baseline	32.34±0.19	30.56±0.44	29.84±0.37	27.65±0.48
(b) Resize & identity	32.47±0.20	30.33±0.17	29.64±0.63	27.62±0.25
(c) Resize & 3x3 conv + 3x3 conv	31.36±0.17	32.93±0.78	36.39±0.31	fail
(d) Pooling & partial 3x3 conv + 3x3 conv	31.50±0.29	30.51±0.46	29.84±0.18	28.71±0.45
(e) Pooling & 3x3 conv + 1x1 conv	31.50±0.12	33.33±0.54	36.73±0.03	fail
(f) Pooling & 3x3 conv + 3x3 conv	31.14±0.30	33.99±0.84	36.20±0.20	fail

the size of 32x32, from the image of 4 pixels padded on each side. Then we randomly flip it horizontally. The original 32x32 test images are evaluated for testing. Both the train and test images are normalized with the mean and standard deviation.

Based on the results of experiments with various resize methods, we select bilinear interpolation considering down-sampling quality and computational time. We pad the increasing dimensions with zeros same as in the baseline method.

C. Results

The baseline method (Fig. 2 (a)) is the original ResNet [1] which uses 1x1 pooling of stride 2 for spatial downsampling and zero-padding for increasing channel dimensions, and identity mapping for all the other shortcuts. We adopt it as the baseline instead of the projection shortcuts of using 1x1 convolutions (named as option B in [1]) as recommended in [1].

Table II and Table III show that the model (b), which uses resize when feature map changes in size and identity mapping otherwise, performs best. The next best performance is model (d), which partially breaks the identity mappings only at layers of conv2_1, conv3_2, and conv4_2 of ResNet (see Table I). Thus, it uses identity mapping relatively more than (c), (e), and (f) and therefore performs better.

Although model (d) shows a slightly better performance than the baseline, it can be observed that as the model deepens, the performance gain gets narrow, and in the end, it gets even worse than the baseline in case of the 110 layer model. This suggests that as the model gets deepened, identity mapping becomes more important.

On the contrary, model (b) shows consistently better performance than the baseline even when the model deepens. In this configuration, we use resizing to better preserve identity mapping instead of zero-padding shortcuts and 1x1 pooling. From these results, it can be argued that maintaining identity mapping by downsampling with resizing is better than the

1x1 pooling and zero-padding shortcut scheme, which is the original way of ResNet.

Model (c), (e), and (f) show relatively worse performance overall, where all identity skip connections are broken. Their performance gets worse as the model deepens as opposed to the original method, and they even failed to converge in case of ResNet110. This shows that the more severe breaking-identity is, the more the performance deteriorates.

V. CONCLUSIONS

We experimentally showed the need for identity mapping by constructing various skip connections with or without identity mapping. Use of resize shows the best performance than the conventional 1x1 pooling with a stride of 2 for downsampling, suggesting that maintaining identity even for the case of feature map dimension change helps in improving the original ResNet.

ACKNOWLEDGMENT

This work was supported by Institute of Information & communications Technology Planning & Evaluation (IITP) grant funded by the Korea government(MSIT) (No.2017-0-00053, A Technology Development of Artificial Intelligence Doctors for Cardiovascular Disease, 50%) and (2019-0-00330, Development of AI Technology for Early Screening of Infant/Child Autism Spectrum Disorders based on Cognition of the Psychological Behavior and Response, 50%)

REFERENCES

- [1] K. He, X. Zhang, S. Ren and J. Sun, “Deep Residual Learning for Image Recognition,” 2016 IEEE Conference on Computer Vision and Pattern Recognition, 2016, pp. 770–778
- [2] K. He, X. Zhang, S. Ren, and J. Sun, “Identity mappings in deep residual networks,” In European Conference on Computer Vision (ECCV), 2016, pp. 630–645
- [3] G. Huang, Z. Liu, L. Van Der Maaten and K. Q. Weinberger, “Densely Connected Convolutional Networks,” 2017 IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 2261–2269

- [4] S. Sun, J. Pang, J. Shi, S. Yi, W. Ouyang, "FishNet: A Versatile Backbone for Image, Region, and Pixel Level Prediction," In NeurIPS, 2018
- [5] C. M. Bishop, "Neural networks for pattern recognition," Oxford university press, 1995.
- [6] B. D. Ripley, "Pattern recognition and neural networks," Cambridge university press, 1996.
- [7] W. Venables and B. Ripley, "Modern applied statistics with s-plus," 1999.
- [8] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," In International Conference on Medical image computing and computer-assisted intervention, 2015, pp. 234–241
- [9] T. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan and S. Belongie, "Feature Pyramid Networks for Object Detection," 2017 IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 936–944
- [10] R. K. Srivastava, K. Greff, and J. Schmidhuber, "Highway networks," arXiv:1505.00387, 2015.
- [11] R. K. Srivastava, K. Greff, and J. Schmidhuber, "Training very deep networks," Advances in Neural Information Processing Systems, pp.2377–2385, 2015
- [12] K. He, X. Zhang, S. Ren and J. Sun, "Delving Deep into Rectifiers: Surpassing Human-Level Performance on ImageNet Classification," 2015 IEEE International Conference on Computer Vision, 2015, pp. 1026–1034
- [13] S. Ioffe, and C. Szegedy, "Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift," Proceedings of the 32nd International Conference on Machine Learning, in Proceedings of Machine Learning Research, vol. 37, pp.448-456, 2015